Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Live Oak H (Formerly Silveira Dairy Inc.)

Physical address of dairy:

Number and Street	City	County	Zip Code
20271 Johnson AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

XX45-X140-XX26-XXXX

B. OPERATORS

Nyman, Brad

Operator name: Nyman, Brad

Telephone no.: (209) 495-8821 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1331	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Silveira, Frank

Legal owner name: Silveira, Frank

Telephone no.: (209) 678-3825 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
20271 Johnson AVE	Hilmar	CA	95324

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	400	200	0	0	0	0
Maximum number	400	200	0	0	0	0
Average number	400	200	0	0	0	0
Avg live weight (lbs)	950	1,000	0	0		

Predominant milk cow breed: Jersey

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 12,349.05 *tons per reporting period*

Total nitrogen from manure: 164,514.20 *lbs per reporting period*

After ammonia losses (30% loss applied): 115,159.94 *lbs per reporting period*

Total phosphorus from manure: 26,400.55 *lbs per reporting period*

Total potassium from manure: 73,975.69 *lbs per reporting period*

Total salt from manure: 234,330.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 26,928,000 *gallons*

Total nitrogen generated: 61,855.58 *lbs*

Total phosphorus generated: 16,691.02 *lbs*

Total potassium generated: 85,989.97 *lbs*

Total salt generated: 507,598.64 *lbs*

	26,928,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	26,928,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1 (Dad's)	17	17	2	process wastewater	XX45-X140-XX24-XXXX XX45-X140-XX25-XXXX
2 (The 25)	24	24	2	process wastewater	XX45-X140-XX27-XXXX
3 (The 30)	28	28	2	process wastewater	XX45-X140-XX25-XXXX XX45-X140-XX26-XXXX
4 (The 60)	57	57	2	process wastewater	XX45-X140-XX21-XXXX XX45-X140-XX24-XXXX
Totals for areas that were used for application	126	126	8		
Totals for areas that were not used for application					
Land application area totals	126	126	8		

B. CROPS AND HARVESTS

1 (Dad's)

Field name: 1 (Dad's)

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	304.92 *ton*	As-is		70.0	4,400.00	1,300.00	6,800.00		7.57

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.94	157.84	46.63	243.94	814.67

1 (Dad's)

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	543.67 *ton*	As-is		68.6	3,300.00	1,000.00	3,100.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.98	211.07	63.96	198.28	1,044.36

2 (The 25)

Field name: 2 (The 25)

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	434.40 *ton*	As-is		70.0	4,200.00	1,100.00	13,000.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.10	152.04	39.82	470.60	488.70

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	758.40 *ton*	As-is		70.2	3,200.00	1,000.00	2,900.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.60	202.24	63.20	183.28	1,035.85

3 (The 30)

Field name: 3 (The 30)

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	512.80 *ton*	As-is		70.0	4,200.00	1,100.00	13,000.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.31	153.84	40.29	476.17	494.49

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	862.40 *ton*	As-is		70.2	3,200.00	1,000.00	2,900.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.80	197.12	61.60	178.64	1,009.62

4 (The 60)

Field name: 4 (The 60)

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	980.40 *ton*	As-is		69.1	5,900.00	1,400.00	15,500.00		7.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.20	202.96	48.16	533.20	831.23

4 (The 60)

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	1,788.03 *ton*	As-is		69.2	3,500.00	900.00	3,700.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.37	219.58	56.46	232.13	966.16

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 (Dad's) - 11/15/2022: Oats, silage-soft dough

Field name: 1 (Dad's)

Crop: Oats, silage-soft dough Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.42	11.72	75.24	319.33	234,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	308.84	2,926,312.00 *gal*
Application event totals		43.42	11.72	75.24	628.17	

01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.54	8.27	48.14	316.68	144,000.00 *gal*
Application event totals		43.54	8.27	48.14	316.68	

02/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.43	10.34	60.17	395.85	180,000.00 *gal*
Application event totals		54.43	10.34	60.17	395.85	

03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.31	12.41	72.21	475.02	216,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	237.01	2,245,705.00 *gal*
Application event totals		65.31	12.41	72.21	712.03	

1 (Dad's) - 06/09/2023: Corn, silage

Field name: 1 (Dad's)

Crop: Corn, silage Plant date: 06/09/2023

1 (Dad's) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.46	15.32	141.50	678.95	612,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	249.01	2,359,423.00 *gal*
Application event totals		44.46	15.32	141.50	927.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	236.10	2,237,040.00 *gal*
Application event totals		0.00	0.00	0.00	236.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.79	27.19	104.86	682.84	828,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	224.67	2,128,807.00 *gal*
Application event totals		87.79	27.19	104.86	907.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	215.53	2,042,198.00 *gal*
Application event totals		0.00	0.00	0.00	215.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.71	14.78	56.99	371.11	450,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	211.74	2,006,249.00 *gal*
Application event totals		47.71	14.78	56.99	582.85	

1 (Dad's) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	198.00	1,876,113.00 *gal*
Application event totals		0.00	0.00	0.00	198.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.54	9.46	36.47	237.51	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	164.45	1,558,179.00 *gal*
Application event totals		30.54	9.46	36.47	401.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.34	23.64	91.19	593.77	720,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	156.64	1,484,211.00 *gal*
Application event totals		76.34	23.64	91.19	750.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	169.54	1,606,416.00 *gal*
Application event totals		0.00	0.00	0.00	169.54	

2 (The 25) - 11/15/2022: Oats, silage-soft dough

Field name: 2 (The 25)

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 (The 25) - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.32	12.77	81.99	347.99	360,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	304.93	4,078,920.00 *gal*
Application event totals		47.32	12.77	81.99	652.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.69	11.72	68.20	448.63	288,000.00 *gal*
Application event totals		61.69	11.72	68.20	448.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.98	10.25	59.67	392.55	252,000.00 *gal*
Application event totals		53.98	10.25	59.67	392.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.26	8.79	51.15	336.47	216,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	241.43	3,229,464.00 *gal*
Application event totals		46.26	8.79	51.15	577.90	

2 (The 25) - 06/09/2023: Corn, silage

Field name: 2 (The 25)

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 (The 25) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.02	17.24	159.19	763.82	972,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	266.81	3,569,064.00 *gal*
Application event totals		50.02	17.24	159.19	1,030.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	236.87	3,168,480.00 *gal*
Application event totals		0.00	0.00	0.00	236.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	222.68	2,978,640.00 *gal*
Application event totals		0.00	0.00	0.00	222.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.59	20.94	80.74	525.74	900,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	195.57	2,616,010.00 *gal*
Application event totals		67.59	20.94	80.74	721.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	181.22	2,424,113.00 *gal*
Application event totals		0.00	0.00	0.00	181.22	

2 (The 25) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.48	18.42	71.05	462.65	792,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	171.15	2,289,416.00 *gal*
Application event totals		59.48	18.42	71.05	633.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.67	15.07	58.13	378.53	648,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	151.55	2,027,196.00 *gal*
Application event totals		48.67	15.07	58.13	530.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	150.18	2,008,946.00 *gal*
Application event totals		0.00	0.00	0.00	150.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.96	14.24	54.90	357.50	612,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	142.67	1,908,416.00 *gal*
Application event totals		45.96	14.24	54.90	500.17	

3 (The 30) - 11/15/2022: Oats, silage-soft dough

Field name: 3 (The 30)

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 (The 30) - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.78	15.32	98.39	417.58	504,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	304.83	4,757,144.00 *gal*
Application event totals		56.78	15.32	98.39	722.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.48	11.30	65.76	432.60	324,000.00 *gal*
Application event totals		59.48	11.30	65.76	432.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.66	7.53	43.84	288.40	216,000.00 *gal*
Application event totals		39.66	7.53	43.84	288.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/21/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.26	8.79	51.15	336.47	252,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	233.78	3,648,372.00 *gal*
Application event totals		46.26	8.79	51.15	570.25	

3 (The 30) - 06/09/2023: Corn, silage

Field name: 3 (The 30)

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 (The 30) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.05	15.87	146.55	703.20	1,044,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	264.92	4,134,368.00 *gal*
Application event totals		46.05	15.87	146.55	968.12	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	239.08	3,731,140.00 *gal*
Application event totals		0.00	0.00	0.00	239.08	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.10	18.30	70.59	459.64	918,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	224.29	3,500,224.00 *gal*
Application event totals		59.10	18.30	70.59	683.93	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	211.52	3,300,916.00 *gal*
Application event totals		0.00	0.00	0.00	211.52	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.16	22.97	88.58	576.81	1,152,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	195.19	3,046,207.00 *gal*
Application event totals		74.16	22.97	88.58	772.00	

3 (The 30) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	189.75	2,961,321.00 *gal*
Application event totals		0.00	0.00	0.00	189.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	177.79	2,774,609.00 *gal*
Application event totals		0.00	0.00	0.00	177.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.30	16.51	63.67	414.58	828,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	160.52	2,505,116.00 *gal*
Application event totals		53.30	16.51	63.67	575.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.40	12.20	47.06	306.43	612,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	155.37	2,424,691.00 *gal*
Application event totals		39.40	12.20	47.06	461.80	

4 (The 60) - 11/15/2022: Oats, silage-soft dough

Field name: 4 (The 60)

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

4 (The 60) - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.80	13.98	89.76	380.95	936,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	286.80	9,111,336.00 *gal*
Application event totals		51.80	13.98	89.76	667.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.19	10.48	61.02	401.40	612,000.00 *gal*
Application event totals		55.19	10.48	61.02	401.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.93	12.33	71.78	472.24	720,000.00 *gal*
Application event totals		64.93	12.33	71.78	472.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.21	8.02	46.66	306.96	468,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	233.56	7,419,975.00 *gal*
Application event totals		42.21	8.02	46.66	540.51	

4 (The 60) - 06/09/2023: Corn, silage

Field name: 4 (The 60)

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

4 (The 60) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.10	12.10	111.71	536.01	1,620,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	261.93	8,321,430.00 *gal*
Application event totals		35.10	12.10	111.71	797.94	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	237.41	7,542,240.00 *gal*
Application event totals		0.00	0.00	0.00	237.41	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.89	20.10	77.51	504.71	2,052,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	222.49	7,068,456.00 *gal*
Application event totals		64.89	20.10	77.51	727.20	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.24	14.63	56.43	367.46	1,494,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	214.86	6,826,113.00 *gal*
Application event totals		47.24	14.63	56.43	582.32	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	208.03	6,608,921.00 *gal*
Application event totals		0.00	0.00	0.00	208.03	

4 (The 60) - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	195.92	6,224,118.00 *gal*
Application event totals		0.00	0.00	0.00	195.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.11	25.74	99.27	646.38	2,628,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	188.88	6,000,716.00 *gal*
Application event totals		83.11	25.74	99.27	835.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.06	17.98	69.35	451.58	1,836,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	183.15	5,818,711.00 *gal*
Application event totals		58.06	17.98	69.35	634.73	

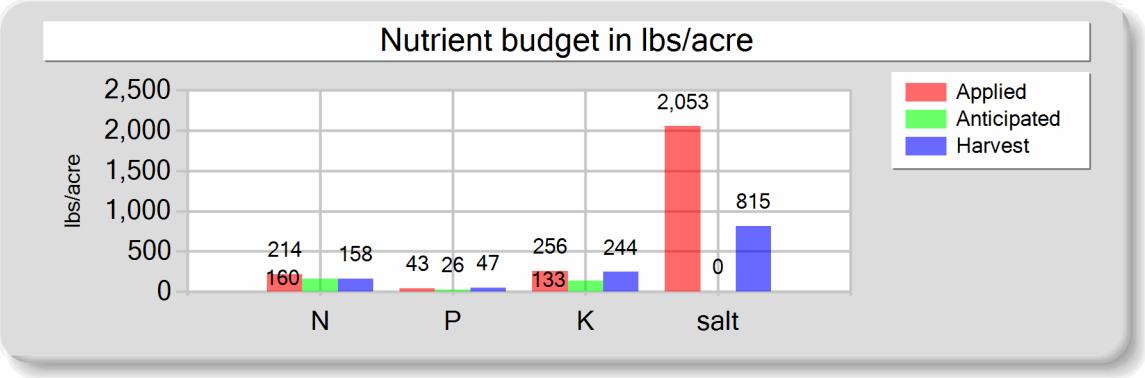
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	179.73	5,709,976.00 *gal*
Application event totals		0.00	0.00	0.00	179.73	

B. NUTRIENT BUDGET

1 (Dad's) - 11/15/2022: Oats, silage-soft dough

Field name: 1 (Dad's) Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.71	42.73	255.75	1,506.87
Fresh water	0.00	0.00	0.00	545.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	213.71	42.73	255.75	2,052.72
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	157.84	46.63	243.94	814.67
Nutrient balance	55.87	-3.90	11.82	1,238.05
Applied to removed ratio	1.35	0.92	1.05	2.52

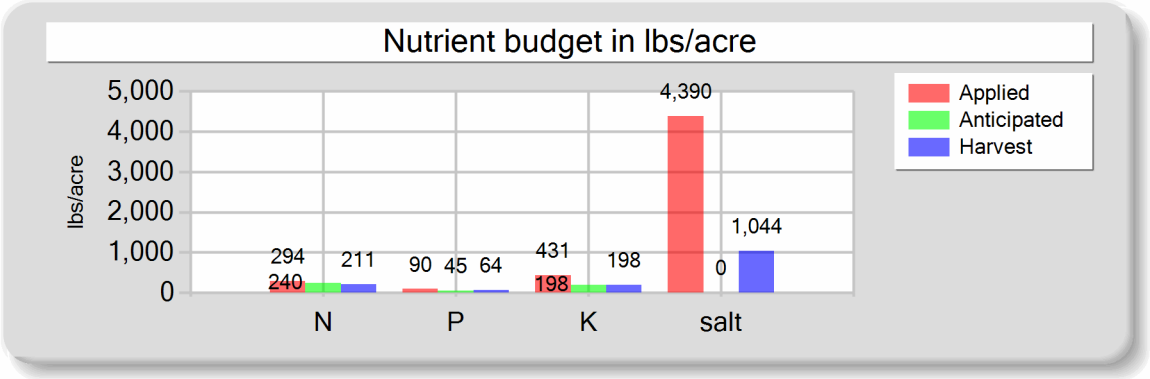
Fresh water applied	
5,172,017.00	*gallons*
190.47	*acre-inches*
11.20	*inches/acre*

Process wastewater applied	
774,000.00	*gallons*
28.50	*acre-inches*
1.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 (Dad's) - 06/09/2023: Corn, silage

Field name: 1 (Dad's) Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	286.85	90.39	431.01	2,564.17
Fresh water	0.00	0.00	0.00	1,825.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.85	90.39	431.01	4,389.87
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	211.07	63.96	198.28	1,044.36
Nutrient balance	82.78	26.43	232.73	3,345.51
Applied to removed ratio	1.39	1.41	2.17	4.20

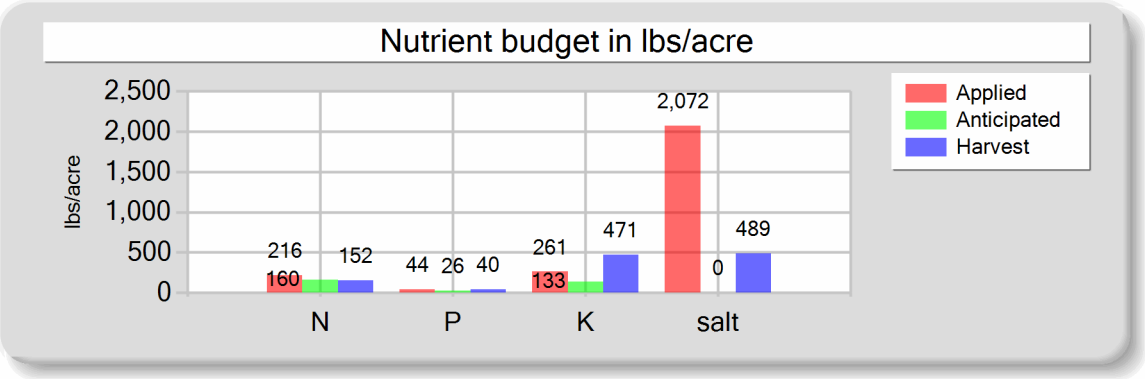
Fresh water applied	
17,298,636.00	*gallons*
637.05	*acre-inches*
37.47	*inches/acre*

Process wastewater applied	
2,898,000.00	*gallons*
106.72	*acre-inches*
6.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 (The 25) - 11/15/2022: Oats, silage-soft dough

Field name: 2 (The 25) Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.24	43.52	261.00	1,525.63
Fresh water	0.00	0.00	0.00	546.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	216.24	43.52	261.00	2,071.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.04	39.82	470.60	488.70
Nutrient balance	64.20	3.70	-209.60	1,583.29
Applied to removed ratio	1.42	1.09	0.55	4.24

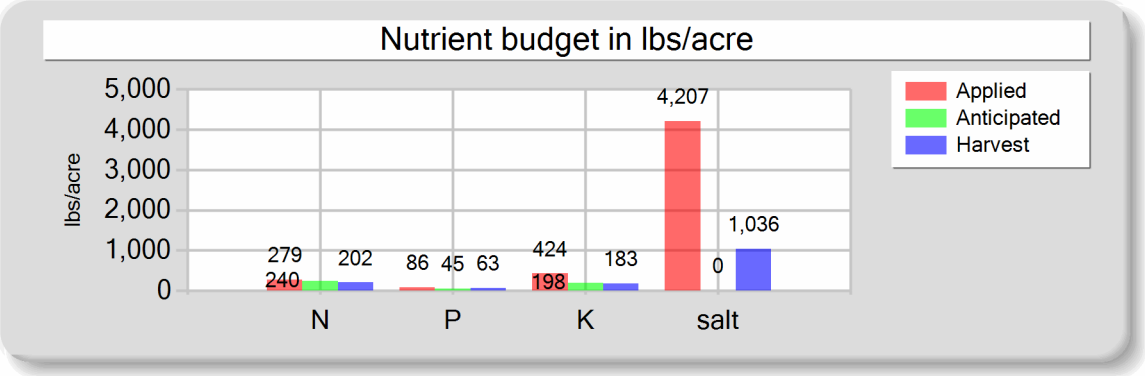
Fresh water applied	
7,308,384.00	*gallons*
269.14	*acre-inches*
11.21	*inches/acre*

Process wastewater applied	
1,116,000.00	*gallons*
41.10	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 (The 25) - 06/09/2023: Corn, silage

Field name: 2 (The 25) Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	271.73	85.91	424.01	2,488.23
Fresh water	0.00	0.00	0.00	1,718.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.73	85.91	424.01	4,206.92
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	202.24	63.20	183.28	1,035.85
Nutrient balance	76.49	22.71	240.73	3,171.07
Applied to removed ratio	1.38	1.36	2.31	4.06

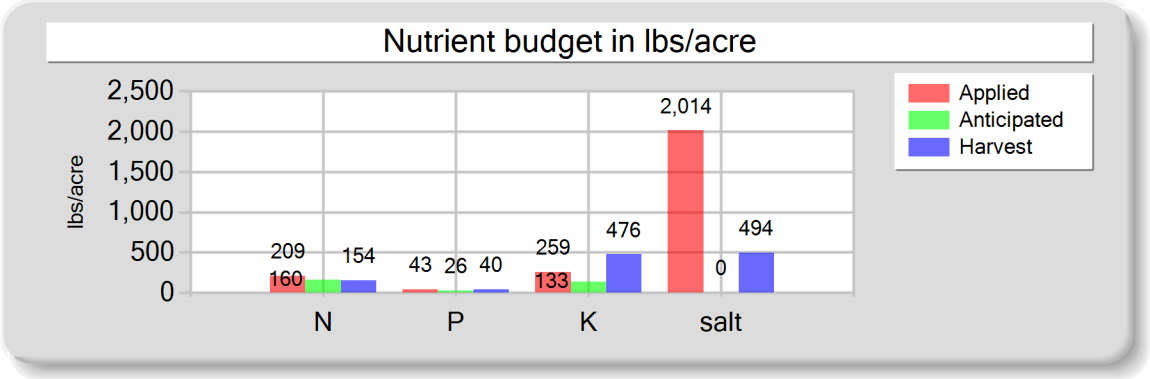
Fresh water applied	
22,990,281.00	*gallons*
846.65	*acre-inches*
35.28	*inches/acre*

Process wastewater applied	
3,924,000.00	*gallons*
144.51	*acre-inches*
6.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 (The 30) - 11/15/2022: Oats, silage-soft dough

Field name: 3 (The 30) Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.18	42.94	259.13	1,475.06
Fresh water	0.00	0.00	0.00	538.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.18	42.94	259.13	2,013.67
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	153.84	40.29	476.17	494.49
Nutrient balance	55.34	2.65	-217.04	1,519.18
Applied to removed ratio	1.36	1.07	0.54	4.07

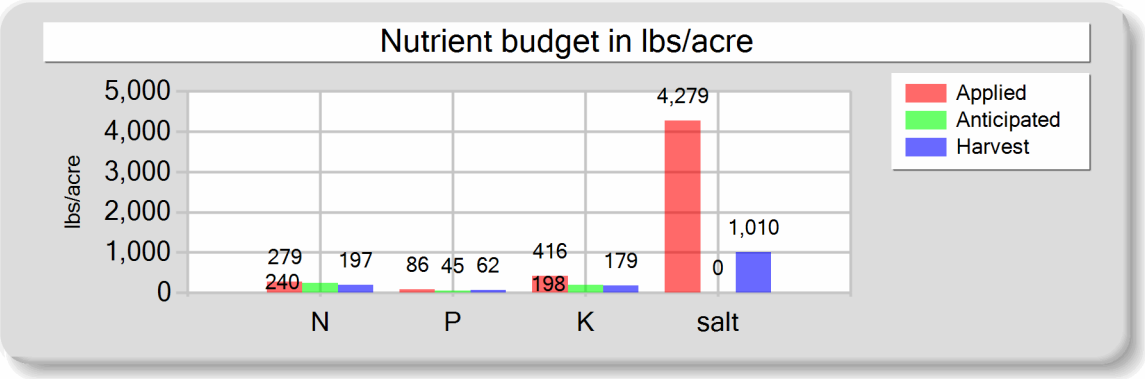
Fresh water applied	
8,405,516.00	*gallons*
309.55	*acre-inches*
11.06	*inches/acre*

Process wastewater applied	
1,296,000.00	*gallons*
47.73	*acre-inches*
1.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 (The 30) - 06/09/2023: Corn, silage

Field name: 3 (The 30) Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	272.01	85.85	416.45	2,460.65
Fresh water	0.00	0.00	0.00	1,818.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.01	85.85	416.45	4,279.09
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	197.12	61.60	178.64	1,009.62
Nutrient balance	81.89	24.25	237.81	3,269.46
Applied to removed ratio	1.42	1.39	2.33	4.24

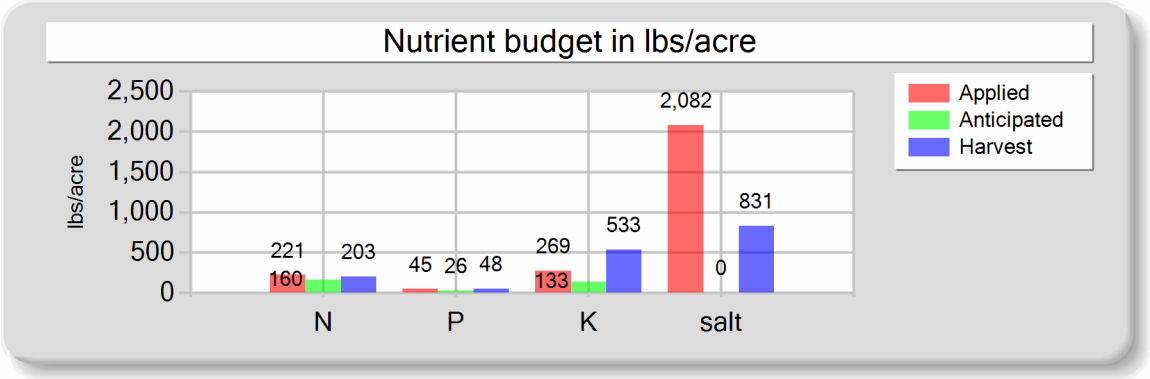
Fresh water applied	
28,378,592.00	*gallons*
1,045.09	*acre-inches*
37.32	*inches/acre*

Process wastewater applied	
4,554,000.00	*gallons*
167.71	*acre-inches*
5.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 (The 60) - 11/15/2022: Oats, silage-soft dough

Field name: 4 (The 60) Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.13	44.81	269.22	1,561.55
Fresh water	0.00	0.00	0.00	520.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.13	44.81	269.22	2,081.90
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	202.96	48.16	533.20	831.23
Nutrient balance	18.17	-3.35	-263.98	1,250.67
Applied to removed ratio	1.09	0.93	0.50	2.50

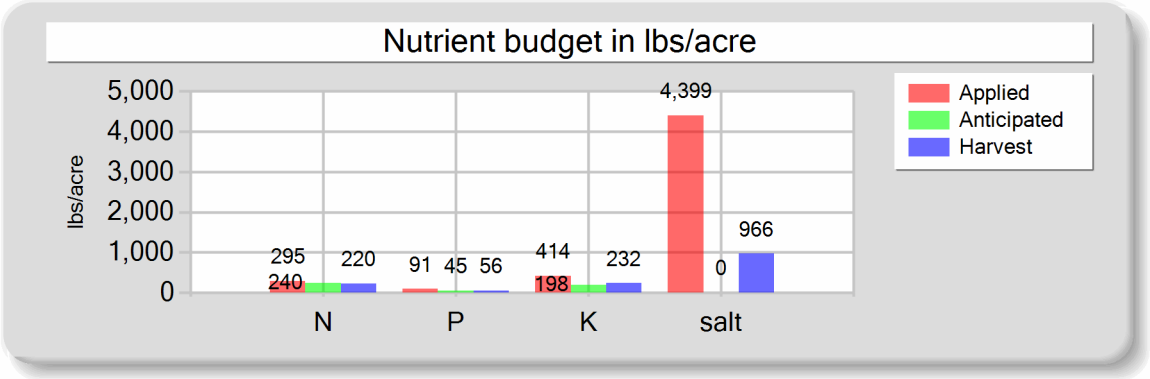
Fresh water applied	
16,531,311.00	*gallons*
608.79	*acre-inches*
10.68	*inches/acre*

Process wastewater applied	
2,736,000.00	*gallons*
100.76	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 (The 60) - 06/09/2023: Corn, silage

Field name: 4 (The 60) Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	288.40	90.55	414.26	2,506.14
Fresh water	0.00	0.00	0.00	1,892.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.40	90.55	414.26	4,398.54
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	219.58	56.46	232.13	966.16
Nutrient balance	75.82	34.08	182.13	3,432.38
Applied to removed ratio	1.35	1.60	1.78	4.55

Fresh water applied	
60,120,681.00	*gallons*
2,214.04	*acre-inches*
38.84	*inches/acre*

Process wastewater applied	
9,630,000.00	*gallons*
354.64	*acre-inches*
6.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Corral Solids

Sample and source description: 2022 Corral Solids

Sample date: 11/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,200.00	5,100.00	4,300.00	17,000.00	5,100.00	800.00	4,400.00	1,020.00		13.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	5,300.00	10,300.00							62.90
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/24/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,500.00	2,900.00	3,200.00	9,200.00	3,500.00	600.00	2,400.00	565.00		4.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	378.00	221.00	0.00	0.00	102.00	655.00								4,110.00	2,780
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	616.00	384.00	0.00	0.00	117.00	681.00								6,160.00	4,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	148.00	106.00	0.00	0.00	51.00	471.00								2,820.00	2,260
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	216.00	123.00	0.00	0.00	66.90	258.00	80.10	57.00	63.00	1,080.00	0.00	115.00	115.00	2,630.00	1,680
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

C. FRESH WATER ANALYSES

Canal

T.I.D Canal

Sample description: T.I.D Canal

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 (Dad's) - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	1,100.00	10,500.00		7.57
DL	100.00	100.00	100.00		0.01

1 (Dad's) - 06/09/2023: Corn, silage

1 (Dad's) - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	1,000.00	3,100.00		5.20
DL	100.00	100.00	100.00		0.01

2 (The 25) - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	900.00	10,400.00		4.50
DL	100.00	100.00	100.00		0.01

2 (The 25) - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	1,000.00	2,900.00		5.50
DL	100.00	100.00	100.00		0.01

3 (The 30) - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	900.00	10,400.00		4.50
DL	100.00	100.00	100.00		0.01

3 (The 30) - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	1,000.00	2,900.00		5.50
DL	100.00	100.00	100.00		0.01

4 (The 60) - 11/15/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,900.00	1,400.00	15,500.00		7.82
DL	100.00	100.00	100.00		0.01

4 (The 60) - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	900.00	3,700.00		5.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

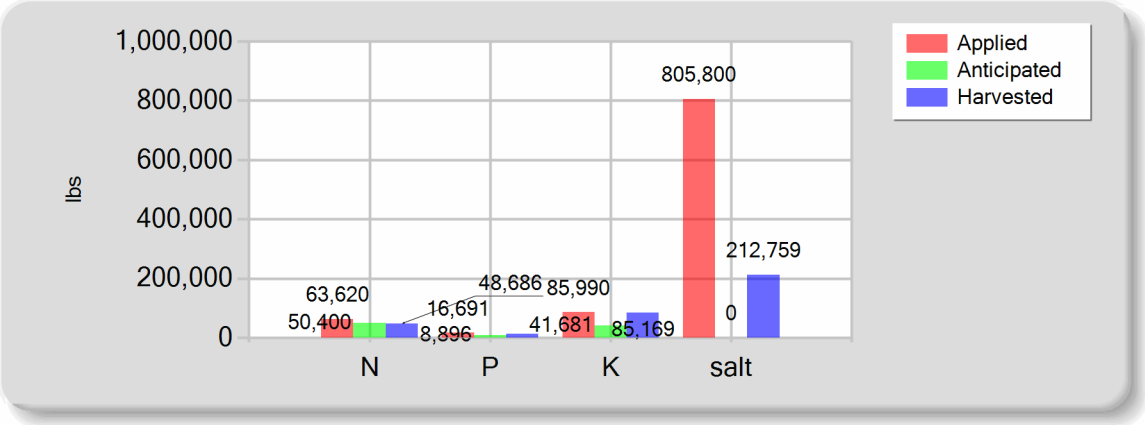
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

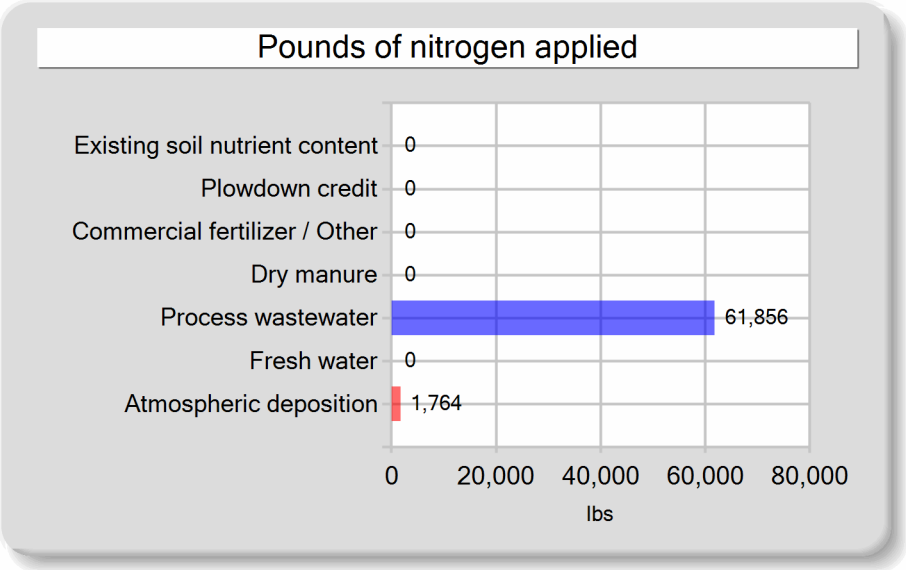
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

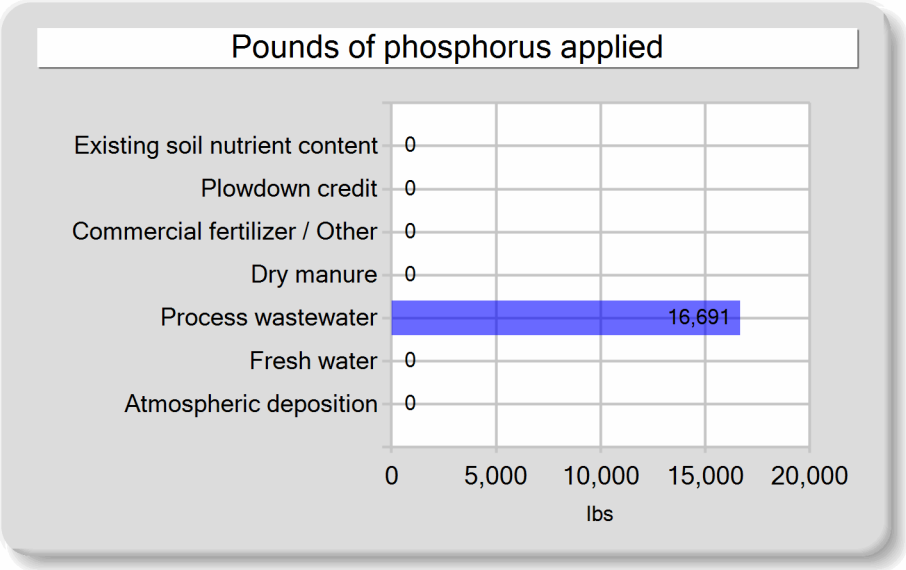
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	61,855.58	16,691.02	85,989.97	507,598.64
Fresh water	0.00	0.00	0.00	298,201.61
Atmospheric deposition	1,764.00	0.00	0.00	0.00
Total nutrients applied	63,619.58	16,691.02	85,989.97	805,800.25
Anticipated crop nutrient removal	50,400.00	8,895.60	41,680.80	0.00
Actual crop nutrient removal	48,686.05	13,169.15	85,169.33	212,759.48
Nutrient balance	14,933.53	3,521.87	820.64	593,040.77
Applied to removed ratio	1.31	1.27	1.01	3.79

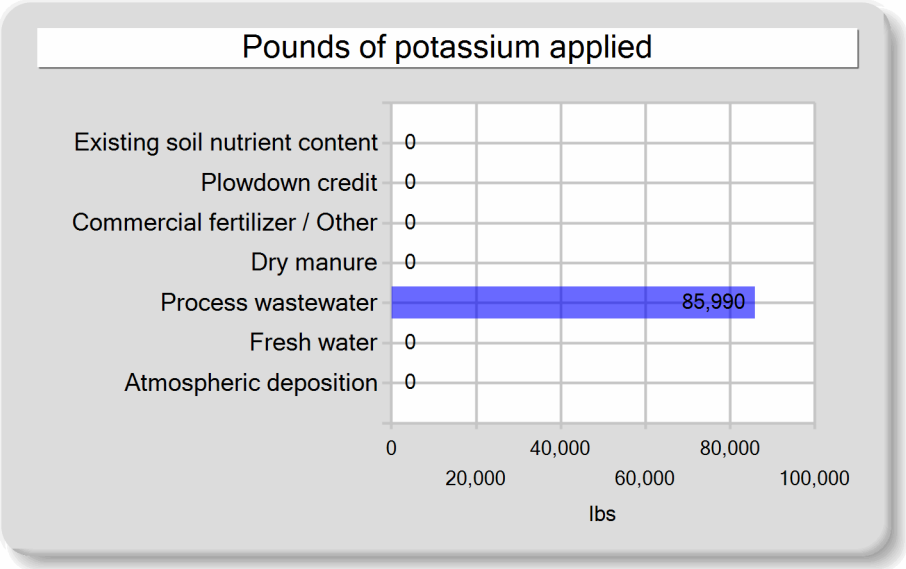
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

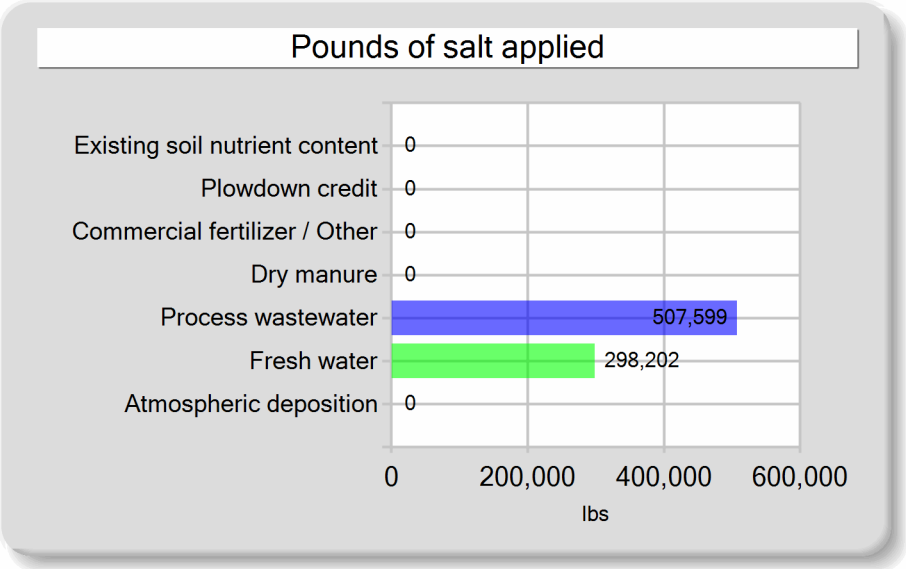


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Silveira	Brad Nyman
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-19-24	6-19-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Live Oak Dairy
P.O. BOX 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: H
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW East	T3240031A	Electrical Conductivity (EC)	EPA 120.1 and	1.06	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	30.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	677	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.8 (T3240030A)	94.6 (T3240030A)	1.26
Electrical Conductivity (EC)	99.9			0.307
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Phone (209) 634-9055 - Fax (209) 634-9057

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Live Oak Dairy
P.O. BOX 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: H
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW West	T3240031B	Electrical Conductivity (EC)	EPA 120.1 and	0.383	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.18	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	245	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.8 (T3240030A)	94.6 (T3240030A)	1.26
Electrical Conductivity (EC)	99.9			0.307
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Live Oak H
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: BN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/24	2:00	Domestic	DW East		16.6'	✓				
2	S	2:15	Wells	DW West		17.8'	✓				
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged in By:	LABORATORY NUMBER
1st		FRAG				
2nd		DENELE ANALYTICAL, INC.	8-28-23 1136			T3240031

Required to be submitted to any agency? (YES / NO)